

VLSI Template Stage Output Matrix

VLSI stage map

Template Stage	Board-Specific Output	Files	Status	Engineer Action
1. Front-End Design	Schematic closure package	04_front_end_schematic, 07_netlists, 16_verification	Release pending: 564 schematic parity items remain open.	Reconcile schematic/netlist/PCB and close or sign-waive every parity item.
2. Back-End Design	PCB ECO package	08_board_outline_mechanical, 10_component_placement, 11_routing_planes	Release pending: 32 DRC violations remain open.	Fix edge clearances, placeholders, mounting-hole clearance, and silkscreen issues.
3. Analog Design	TFLN optical/RF/bias review	04_front_end_schematic, 06_footprints_symbols, 12_silkscreen_drc_fab_notes	Review output issued.	Validate U3 TFLN PIC, U50-U53 RF drivers, bias rails, monitor lines, and MZM lanes.
4. Mixed-Signal Design	NCE-to-TFLN interface review	04_front_end_schematic, 07_netlists	Review output issued.	Confirm TFLN_TX lane mapping plus SPI/I2C/JTAG/PLL/AST2600 service interfaces.
5. Verification Domain	Validation reports and issue logs	16_verification	Release pending: DRC/parity open.	Target DRC 0, unconnected 0, parity closed or signed-waived.
6. Physical Verification	DRC, board outline, mechanical, fab notes	08_board_outline_mechanical, 12_silkscreen_drc_fab_notes, 16_verification	Release pending.	Rerun KiCad DRC after ECO and document any mechanical-rule waiver.
7. FPGA Design	Not applicable / interface reserve	02_vlsi_domain_outputs/FPGA_DOMAIN_NOTE	No FPGA-specific RTL supplied in the current package.	If FPGA/programmable logic is added, issue RTL, constraints, bitstream, and verification logs.
8. Embedded Systems	AST2600/service-interface checklist	02_vlsi_domain_outputs/EMBEDDED_SYSTEMS_SERVICE_CHECKLIST	Checklist output issued.	Confirm firmware ownership, boot straps, management bus, sensors, reset and update path.
9. CAD / EDA Tool Development	KiCad project, generated reports, manifest	00_OPEN_IN_KICAD, FULL_PACKAGE_MANIFEST	Generated in KiCad-compatible handoff format.	Use KiCad 10.0.3 or later and preserve generated checksum manifest.
10. Semiconductor Fabrication	PCB fab/assembly review package	13_gerbbers_drill, 14_dfa_assembly, 15_dfm_manufacturability, 17_release_gate	DFM/DFA review package issued; production release follows closure.	Complete DFM/DFA and regenerate final production outputs after ECO closure.